

# Supplementary material

## Correction-Aware Data Admission: Candidate-Bound Authorization and Field-Level Provenance for AI-Derived Updates

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This supplement documents the evidence boundaries, formal and implementation checks, a worked admission example, complete current-version timing grids, and historical integration evidence. Sections, tables, and figures prefixed S belong to this document. No new experiment is implied by reformatting archived results here.

### **S1. Artifact, protocol, and reproduction**

#### *S1.1. Three evidence layers*

The reference implementation is fixed at commit `c6d512843c905cab6d8521dd8c914f7fb26d85ae` of <https://github.com/lhh666-6/auto-dete>, using `latest/code/implementation-fixed`. The tag `r31-jss-2026-09-13` retains formal models, observation witnesses, repaired code, and historical analyses. The historical `v8` evidence remains under `r21-jss-2026-09-07-v8`. These layers are not substituted for one another when reporting counts or timings.

The new E1–E3 package is fixed at <https://github.com/lhh666-6/auto-dete/tree/2645e5e18c900ea91c9c980e44195dc71e410432/DK-E-supplement>. Its `PROTOCOL.md` was drafted before outcome collection on 23 September 2026 and records pilot-driven amendments. It is a local prospective execution plan, not an externally registered protocol. Pilots remain in separately named directories and are excluded from final statistics. No hosted-model call was made in E1–E3.

#### *S1.2. Inputs, cases, and negative controls*

E1 selects the first lexicographically ordered archived run containing a proposal in each of the G1/G2/D1 by B1–B4 cells; success labels are not a selection criterion. The twelve cells contain four distinct proposed values: 8, 100, the string B-008, and 9. Three additional controls exercise

Table S1: New experiment evidence and interpretation units. Paths are relative to the pinned DKE-supplement directory.

| Layer     | Primary records                                                                                                 | Unit and limit                                                                |
|-----------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| E1        | <code>results/e1/inputs.json</code> ;<br><code>results/e1/receipts.jsonl</code>                                 | 495 constructed executions; 165 per arm; seven distinct proposal values.      |
| E2        | <code>results/e2/browser-observations.jsonl</code> ; screenshots and databases                                  | 60 scripted cases plus six replay setup confirmations; not a human study.     |
| E3        | <code>results/e3/*/raw.jsonl</code> ;<br><code>summary.json</code> ; <code>paired.json</code>                   | 22,400 timed observations, excluding warmups; one machine.                    |
| Checks    | <code>rejection-audit.json</code> ;<br><code>verification.json</code> ;<br><code>regression-receipt.json</code> | Expected rejection causes, counts, state digests, and test execution receipt. |
| Integrity | <code>artifact-manifest.json</code> ;<br><code>source-integrity.json</code>                                     | Content identities, not third-party attestation.                              |

$2 \rightarrow 2.0$ ,  $1 \rightarrow \text{true}$ , and nested integer/float changes. These values enter newly constructed form histories with fresh certificates and review records. This is not an end-to-end replay of the original tasks. Input records retain source paths, event identifiers, and hashes.

The eleven cases per input are Accept, Correction, multi-field admission, equal-valued candidate substitution, different-evidence substitution, stale authorization, replay, an unauthorized submitted value, and interruptions after CAS, before transition insertion, and before commit. The rich-context journal retains field, version, evidence, producer and proposal context, but its review projection excludes certificate identity, candidate identifier, and instance creation time. Its exact variant adds the reviewed-candidate link. Each successful three-field record is checked with six queries per field: proposed value, authorized value, reviewer, reviewed candidate, current value, and source version. The 18 answers per accepted case are query observations, not independent experiments.

E2 uses the JSON values 100, 2.5, and true. Each appears in ten interactions per path: Accept, Correction, legitimate reselection, candidate-request substitution, authorized-value substitution, principal substitution, stale state, replay, precommit interruption, and display-only substitution. The driver records DOM candidate identity, proposal, and authorized value before confirmation. Replay cases each require an initial confirmation, explaining the

six additional setup calls. The float value 2.5 avoids JavaScript serialization normalizing an integral float to an integer. The gate records a server-held review session using fixed simulated principal labels and a database separate from the fact database.

The rejection audit checks all 372 rejected attempts across E1 and E2, including their exact expected error categories or injected failpoints. No rejected attempt changes the measured fact-database digest. The baseline is taken after authorization preparation: already recorded review intent is not required to disappear when admission fails. This distinction also prevents the session database from being silently included in a fact-database atomicity claim.

### *S1.3. Execution environment and commands*

The recorded environment is Intel Core i7-10875H, Windows 10 build 19045, CPython 3.11.16, SQLite 3.53.1, and SQLAlchemy 2.0.51. Browser orchestration uses Playwright and local Chrome. E3 uses a seeded random ordering of the two arms (seed 20260923), five warmups, and 200 timed observations per arm and cell. Admission uses SQLite DELETE/FULL; the storage-population measurement uses WAL/NORMAL followed by check-pointing. Background operating-system activity was not controlled.

The deposited README specifies dependency installation, reference-source resolution, and the separate browser driver environment. The following commands illustrate the entry points; the output directory must be fresh, and the reference checkout must be configured as described there. Performance should run after other tests have finished.

```
python -B experiment1.py -output results/replicate-e1
python -B experiment2.py -output results/replicate-e2
python -B experiment3.py -output results/replicate-e3
```

Pilot flags reduce coverage and cannot reproduce final denominators. The reference regression invocation covers its value-equality integration tests and conformance directory (85 passes); the supplementary test module records ten passes. The regression receipt transcribes the execution tool output rather than presenting itself as a pytest-generated JUnit report. The manifest records 61 pre-run measurement-source hashes; the reporting check found them unchanged. The original artifact manifest covers 3,068 files.

#### *S1.4. Lossless storage packaging*

The 100,000-transition full database is 141,062,144 bytes and is published as `results/e3/storage/storage-100000/full.db.gz`. The publication package includes `restore_large_database.py` and `publication-packaging.json`. Restoration checks the uncompressed SHA-256 against the original manifest; its verified value is

```
a6ab45b86b3f198e8452b7adae756012807efe09ceb92d32c759cd9d
2ffb816e.
```

Compression changes transport size, not the database footprint used in the storage result. The restoration helper refuses to overwrite a differing existing file. Source and artifact manifests support reproducibility checks; they do not establish external validity or independent replication.

## **S2. Formal and implementation evidence**

### *S2.1. Complete observation functions and witness construction*

The main text retains the observation basis, opaque-handle condition, outcome definition, proposition, and five-witness proof. The following table and relational representation specify the complete executable projection.

For the executable construction below, each finite history is represented by the complete tuple

$$\hat{h} = \langle r, C, A, B, v_{\text{exp}}, v_{\text{obs}}, F_{\text{decl}}, K, V_v, V_{v'}, S_v, S_{v'}, T \rangle, \quad (1)$$

containing the target, candidate and authorization registries, batch references, expected and observed pre-versions, declared fields, a commit sequence  $K$ , endpoint value and source maps, and a registry  $T$  of source-transition identifiers, fields, and values. Each commit step records a distinct successor identity and complete before/after value maps. The domain predicate checks unique candidate and authorization keys, resolvable batch references, schema-valid fields, total value maps, and a connected commit sequence from  $V_v$  to  $V_{v'}$ . Its committed effect domain is the union of the steps' actual value-change domains; its successor count is  $|K|$ . Neither is an independently assigned witness attribute.

Source observations expose each field's anchor count, reverse-resolution count in  $T$ , agreement of the resolved field/value with the successor, and exact prior-source retention for an unchanged field. Missing, duplicate, or

Table S2: Observation functions used by the conditional characterization. “Derives” lists information available without consulting another class.

| Class | Raw history components exposed                                                                                | Derives                                                                              | Deliberately excludes                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| $D_C$ | opaque candidate and authorization-target handles; candidate and admission targets; field, evidence, producer | candidate binding and non-temporal context agreement                                 | decoded proposal/authorized values, current version, batch-effect domain, successor sources |
| $D_V$ | proposal value, authorized value, committed value, and their role labels for an anonymous batch item          | Accept versus Correction and proposal/authority attribution                          | candidate handle and context, temporal order, batch completeness, source identities         |
| $D_F$ | expected and transactionally observed pre-state discriminator at the admission attempt                        | current versus superseded authorization                                              | proposal roles, non-temporal candidate context, batch effects, source relation              |
| $D_B$ | declared item-field domain, committed effect domain, commit grouping, and successor count                     | complete versus partial/fragmented successor construction                            | value-role attribution, freshness, and field-source identities                              |
| $D_S$ | successor field-source relation; local resolution and copy-forward checks                                     | missing/ambiguous sources, field/value inconsistency, and replaced unchanged sources | commit grouping, value roles, freshness, and candidate context                              |

inconsistent sources remain semantic failures rather than being excluded by the domain predicate. These are local source checks: the complete authorization–candidate–evidence chain belongs to the main text’s operational P6 checks. Source observations hide intermediate commit grouping; freshness observes the admission-entry pre-state comparison. These explicit observation boundaries make it possible to compare atomic and fragmented histories with identical endpoints.

The main text retains Proposition 1, its five-witness proof, and the explicit reduced-projection equation. The original candidate/context witness changes record context with a fixed candidate handle. The separate identity-isolation pair fixes the two-candidate registry and the non-identity observations, then

changes the admitted candidate. The executable checker derives outcomes and observations from the complete tuple rather than storing either as an independent attribute.

### *S2.2. Evidence identity and operational enforcement details*

Evidence identity is not content equality alone. The realization uses

$$EID(e) = \langle H(\text{bytes}(e)), L(r, f, u, \kappa) \rangle, \quad (2)$$

where  $H$  is SHA-256 and  $L$  is a versioned canonical locator containing the owner record, related field, normalized storage-relative URI  $u$ , and optional crop identity  $\kappa$ . The Alloy model abstracts content and locator as primitive domains. Its `certIdentityFacts` assumes unique certificate identities over the modeled certificate objects, excluding identity collisions from the checked scopes. Relating this abstraction to the implementation requires collision resistance of the deployed digest and integrity of the canonicalization and evidence store; it is not a proof or claim that SHA-256 is mathematically injective.

Operational properties P0–P6 and their interpretation remain in the main contract table.

*Authorization timing and revocation.* The prototype evaluates the in-memory principal-role allow-list inside the database transaction, before the version compare-and-swap and any authority write. A revocation that completes before this check causes zero-write rejection. The allow-list lock protects only the policy lookup/update; it is not held for the remainder of the database transaction. Consequently, a revocation that overlaps an admission after its successful policy check does not cancel that in-flight commit. The implementation therefore provides commit-entry revalidation, not linearizable revocation against already admitted operations, distributed identity-provider semantics, or session invalidation. Deployments requiring those properties must place a transactional policy epoch or equivalent revocation discriminator inside the admission compare-and-swap.

Human authorization is not a truth oracle. The contract establishes who authorized which persisted candidate and value in which context, and whether the successor was constructed atomically. It cannot establish that the authorized value is factually correct.

Table S3: Mapping from failure-distinguishing information classes to bounded and concrete evidence.

| Class | Safe/unsafe witness                                                                                 | Alloy support                                                                                                                              | Concrete support                                                                      |
|-------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| $D_C$ | matching / mismatched non-temporal candidate context; separate equal-valued identity-isolation pair | field, evidence-identity, evidence-owner, record, authorization-certificate, certificate-preexistence, and evidence-preexistence ablations | record, field, evidence, and certificate substitution cases                           |
| $D_V$ | retained Correction / wrong attribution of the same final value                                     | legal Correction witnesses and authorization-value ablation                                                                                | singleton, all-field, and mixed Correction lifecycles with immutable candidate checks |
| $D_F$ | current / superseded authorization                                                                  | certificate-version and freshness ablations; stale-rejection assertions                                                                    | stale replay, contending confirmation, and compare-and-swap rejection                 |
| $D_B$ | atomic / partial or fragmented successor                                                            | transition-bijection ablation and whole-batch framing assertions                                                                           | exact changed-field validation, failpoints, rollback, and concurrency cases           |
| $D_S$ | complete / missing, replaced, or duplicate source                                                   | source-anchor attack witnesses and P6 trace assertions                                                                                     | source-corruption catalogue and reverse-trace validation                              |

### *S2.3. Information-class and persisted-state mappings*

### *S2.4. Legal, unsafe, and preservation commands*

Six legal-witness commands cover multi-field Accept, Correction, mixed Accept/Correction, same-value admission, initial snapshot, and singleton admission. Eleven paired ablations remove the conjuncts mapped in Table S3. Principal membership remains fixed; the unused `ABL_7` marker is excluded because host identity trust is an external assumption.

Three source-attack commands start from a legal Full prefix and then remove a source anchor, replace it, or add a duplicate source-version transition; each requires an incomplete trace. Fourteen assertions run per scope. Nine guard definitions and immediate consequences: P0/P1/P3/P4/P5, whole-batch effects, rejection framing, and two P6 checks. Five check invariant preservation, two singleton correspondences, rejection of an ablated attempt, and positive trace completeness under a well-formed pre-state.

Table S4: Principal elements of the executable abstraction.

| Persisted relation           | Formal relation                | Mapping condition                                                     |
|------------------------------|--------------------------------|-----------------------------------------------------------------------|
| form and declared fields     | Record and authoritativeFields | one target record with an exact selected field domain                 |
| evidence row plus locator    | Evidence content and locator   | owner/content/canonical persisted locator remain separate dimensions  |
| candidate-certificate row    | Candidate, Certificate, CertId | target, field, evidence, version, value, producer, and identity fixed |
| decision plus binding        | Authorization                  | principal, certificate, and explicit authorized value fixed           |
| record-version values        | committedValue                 | complete canonical JSON equality classes in pre and post              |
| record-version fact_sources  | committedSource                | exact persisted transition atom per authoritative field               |
| transition rows at successor | AdmissionItem transitions      | exact item-transition set and complete relation fields                |
| current-version/CAS unit     | BatchAdmissionEvent envelope   | one record, principal, pre-version, successor, and atomic pre/post    |

The positive assertion requires a trace-complete predecessor and no stray transition to the successor version; otherwise the permissive base effect admits a post-state that fails **traceComplete**. It covers admitted fields in one step, not arbitrary histories or every carried-forward field. Both profiles include reachable initial and non-vacuous carry-forward witnesses for the precondition. Removing the post-state certificate binding changes UNSAT to SAT. These are separate checks of reachability, bounded preservation, and sensitivity, rather than fourteen independent proofs. Concrete enforcement additionally validates predecessor prefixes and uses transition uniqueness and in-transaction CAS revalidation.

### *S2.5. Scopes and interpretation*

The S1 profile bounds principal structural domains at two records, three fields, six versions, up to six admission items/transitions for most commands, and four states/events. S2 increases representative bounds to three records, four fields, eight versions, up to eight admission items/transitions, and five states/events. Individual commands adjust Value, Evidence, or Transition bounds when a witness needs an extra atom. The artifact contains the executable scopes; this summary is not a replacement for them.

An UNSAT assertion means that the Analyzer found no counterexample within the encoded command scope. A SAT run means that it found the requested witness; the main text cites the Analyzer and solver references. Singleton contract/effect checks additionally guard the batch lift against silently changing the historical one-item semantics.

### *S2.6. Observation witnesses and finite scopes*

The main text defines the observation basis and projection assumptions before stating the omission proposition; the complete functions and construction are given here. The executable witness checker reads complete relational histories and derives both projections and outcome labels; labels are not stored as independent witness attributes. Five class pairs test the original construction. A further pair isolates identity binding with a fixed two-candidate registry and identical non-identity observations; a separate legal control checks single-field change with another field copied forward. The source is `latest/code/formal-fixed/observation_witnesses.py`; the derived report is `latest/evidence/r27-witness/observation-witness-report-v5.json`.

Each Alloy profile contains six legal-witness commands, eleven conjunct-removal witnesses, three source-attack witnesses, two witnesses to the well-formed predecessor condition, and fourteen assertion checks. All 72 outcomes across two profiles match the manifest: 22 SAT and 14 UNSAT per profile. The smaller profile uses representative bounds of two records, three fields, six versions, and four states/events; the larger uses three records, four fields, eight versions, and five states/events. Individual commands adjust other domains. Executable bounds are recorded with the commands in the pinned artifact and should be consulted when reproducing any particular assertion.

The positive trace assertion requires a trace-complete predecessor and no stray transition to the successor version. Nonempty initial and carry-forward witnesses establish reachability of that premise. Removing the post-state

certificate-binding check changes the relevant outcome from UNSAT to SAT. This is bounded sensitivity evidence for the encoded relation, not a proof of arbitrary-history completeness. The formal tamper event alters transitions while freezing values and source pointers; the implementation catalogue also exercises corruption outside that event’s scope.

### *S2.7. Projection and database oracles*

The projection adapter reads persisted pre/post state independently of the production planner and trace builder, maps JSON equality classes and identity relations into a fixed Alloy instance, and checks committed or rejected wrappers. Nine intended instances are SAT and twenty selected mapping mutants are UNSAT. A rejected wrapper additionally requires equal full-database digests. The separate 35-case catalogue has five legal cases and thirty rejection, corruption, schema-prevention, stateful-evolution, or oracle-mutant cases. Post-persistence corruption requires a changed digest and a trace diagnosis rather than zero writes.

Historical test snapshots have different counts: the initial reported suite has 354 passes, subsequent development has 377, and the r26 isolated export has 389, including optional-Git and revocation checks. These numbers describe their archived invocations. The new 85-test invocation is a selected regression run against the repaired reference source, not a claim to have rerun all 389 checks or to add 85 independent faults. Static-check and historical concurrency receipts remain in the original evidence layer.

### *S2.8. Canonical equality and witness-domain repairs*

An earlier implementation inherited host-language equality that equated integer 2 with float 2.0 and integer 1 with Boolean true. A serialized value could change without obtaining a corresponding changed-field transition. The repair aligns the planner and reverse trace with canonical JSON equality already used by the independent oracle and formal projection. The resulting implementation consistently uses sorted-key, whitespace-minimal JSON serialization for change and value checks. Thirteen targeted regressions include unchanged controls, numeric/Boolean and nested changes, and no-op rejection; seven fail before the repair and all pass afterward. E1 adds explicit type controls using the repaired path.

The witness checker also separates the record schema domain from the declared item domain, allowing a legal change to one field while copying another field’s source forward. This preserves the omission constructions

without requiring every admission to change every field. These repairs and their receipts are versioned separately from v8; the main text’s new cost tables measure the repaired source.

### S3. Worked correction and implementation inspection

Consider a two-field record at version  $v$ : quantity is 90 with source  $t_q$ , and batch is B-008 with source  $t_b$ . Candidate  $c_A$  proposes quantity 100 for version  $v$ . A reviewer authorizes quantity 101 from  $c_A$ . The legal successor  $v + 1$  has quantity 101 from a new transition  $t_A$ , batch B-008 from the exact previous source  $t_b$ , and an authorization naming both  $c_A$  and 101. Candidate  $c_A$  still contains 100. This is an explanatory construction, not an additional measured case.

An equal-valued retry  $c_B$  can have the same value and retained review context while being a different persisted instance. Under instance-level authorization, substituting  $c_B$  for  $c_A$  must fail; under the declared context-level policy it may be admitted. A concurrent successor invalidates the expected predecessor. A proposed two-field admission cannot be split into two commits merely because the final endpoint is the same. Replacing the unchanged batch source with an equal-valued transition also loses the exact source relation.

The following checklist locates the enforcement and observation points. It is an aid derived from the contract, not an independently validated development methodology.

The earlier SQLite demonstrator and its strengthened transactional variant agree with candidate-bound admission on seven of eight single-history cases. Equal-valued substitution is their only decision difference; a ninth paired-history check concerns distinguishability rather than a further admission category. The continuity rerun is deposited in `results/original-baseline-continuity`. It is separate from E1’s richer-context three-arm design and is not added to the 495-execution denominator. In E1, extending the new journal with exact binding eliminates the observed decision and query-answer gap on the tested cases.

Table S5: Design checklist derived from C1–C2. Adapt the probes to the target implementation and retain persisted evidence for each outcome.

| Class | Retain and check                                                                                                                               | Failure probe and expected evidence                                                                                                                     |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| $D_C$ | Exact candidate, target, evidence and producer; bind the decision at review and recheck context inside admission.                              | Substitute an equal-valued candidate from another context: reject without authority writes. Reload the binding to verify candidate identity.            |
| $D_V$ | Proposal, human-authorized value and committed value with separate roles; preserve the candidate and check the latter two values at admission. | Authorize 101 from proposal 100: commit 101, retain 100 and its identity. Reconstruction must expose false machine attribution.                         |
| $D_F$ | Expected and observed predecessor discriminator; compare within the transaction and protect it against races.                                  | Supersede the predecessor before replay: reject without changing the new current state. Competing admissions from one predecessor must not both commit. |
| $D_B$ | Admitted field domain, actual effects and commit grouping; validate the planned successor and commit it atomically.                            | Change two fields; interrupt between writes. Recovery must expose all or none of that admission, never an intermediate or fragmented successor.         |
| $D_S$ | One resolving source per successor field; validate changed-field sources and exact unchanged-source retention before commit and on trace.      | Keep values fixed but remove, duplicate or replace a source. Reject invalid construction; diagnose incomplete trace after persisted corruption.         |

## S4. Complete current-version cost grids

### S4.1. Admission and materialization

The mechanism and materialization experiments each contain ten cells, two arms, and 200 observations per arm/cell: 4,000 timed observations each. Warmups and database-copy setup are excluded. The mechanism timer covers confirmation after adapter construction; the journal opens its sqlite3 connection in construction, whereas the reference ORM can connect lazily inside the call. Different schemas, certificate work, application features, and connection lifecycles confound attribution to an individual integrity check.

Table S6: Complete confirmation paths with pre-registered recognition candidates. Entries are p50 / p95 in milliseconds; 200 observations per arm/cell.

| Fields | Changed | Reference        | Exact journal |
|--------|---------|------------------|---------------|
| 1      | 1       | 28.79 / 50.50    | 10.40 / 13.32 |
| 8      | 1       | 33.99 / 55.40    | 11.19 / 15.18 |
| 8      | 4       | 57.97 / 103.89   | 11.88 / 16.47 |
| 8      | 8       | 75.71 / 129.14   | 21.51 / 26.31 |
| 32     | 1       | 34.75 / 54.66    | 11.41 / 15.90 |
| 32     | 4       | 55.32 / 101.11   | 23.89 / 31.17 |
| 32     | 32      | 187.24 / 292.72  | 13.07 / 17.77 |
| 128    | 1       | 29.35 / 57.56    | 11.33 / 14.76 |
| 128    | 4       | 44.39 / 72.72    | 11.61 / 16.75 |
| 128    | 128     | 712.00 / 1039.83 | 22.09 / 30.53 |

The materialization experiment constructs a valid relational delta before timing, persists that delta in its reduced arm, and checks the final relational fingerprint after timing. The full arm uses manual-source fixtures, whereas the mechanism comparison uses pre-registered recognition candidates. Thus the two full-confirmation columns belong to different workloads and must not be subtracted across experiments. Random identifiers and timestamps also prevent a claim that paired database files are byte-identical; equivalence checks use the declared relational expectations.

### S4.2. Reverse-trace reconstruction

The trace grid crosses fields {1, 8, 32, 128}, versions {1, 10, 100}, and total records {1, 100, 1000}, giving 36 cells and 14,400 timed observations across two arms. Valid unrelated records are seeded before timing. The point-access

Table S7: Full confirmation and prevalidated materialization. Entries are p50 / p95 in milliseconds; 200 observations per arm/cell.

| Fields | Changed | Full confirmation | Materialization |
|--------|---------|-------------------|-----------------|
| 1      | 1       | 23.23 / 33.76     | 9.97 / 14.63    |
| 8      | 1       | 23.87 / 65.83     | 10.40 / 15.05   |
| 8      | 4       | 36.68 / 57.24     | 11.48 / 19.46   |
| 8      | 8       | 50.23 / 97.50     | 11.38 / 17.22   |
| 32     | 1       | 27.00 / 45.75     | 11.00 / 16.52   |
| 32     | 4       | 44.23 / 97.98     | 17.31 / 23.66   |
| 32     | 32      | 122.68 / 200.00   | 13.89 / 22.61   |
| 128    | 1       | 29.10 / 48.08     | 11.08 / 17.21   |
| 128    | 4       | 38.33 / 87.89     | 11.34 / 17.19   |
| 128    | 128     | 427.59 / 621.20   | 23.50 / 37.43   |

arm enumerates identifiers and performs real point loads for the certificate, certificate-evidence, and transition repositories. Both arms run the same current verifier. Equality of the complete trace object is checked after each timed call.

Table S8: Complete current trace grid. Latency is p50 / p95 in milliseconds; each arm has 200 timed observations after five warmups. SQL counts are constant within each arm/cell.

| Fields | Versions | Records | Bulk access   | Point access    | Bulk SQL | Point SQL |
|--------|----------|---------|---------------|-----------------|----------|-----------|
| 1      | 1        | 1       | 6.06 / 8.38   | 7.17 / 10.58    | 12       | 15        |
| 1      | 1        | 100     | 6.15 / 9.27   | 7.33 / 10.88    | 12       | 15        |
| 1      | 1        | 1000    | 7.27 / 11.24  | 8.71 / 13.25    | 12       | 15        |
| 1      | 10       | 1       | 8.12 / 14.57  | 22.44 / 35.33   | 12       | 42        |
| 1      | 10       | 100     | 7.45 / 12.12  | 20.65 / 32.48   | 12       | 42        |
| 1      | 10       | 1000    | 7.85 / 12.69  | 21.72 / 34.71   | 12       | 42        |
| 1      | 100      | 1       | 20.52 / 24.14 | 152.07 / 163.52 | 12       | 312       |
| 1      | 100      | 100     | 22.35 / 36.98 | 164.96 / 205.99 | 12       | 312       |
| 1      | 100      | 1000    | 24.59 / 38.62 | 177.42 / 219.39 | 12       | 312       |
| 8      | 1        | 1       | 7.49 / 10.93  | 18.32 / 27.82   | 12       | 36        |
| 8      | 1        | 100     | 8.12 / 14.26  | 19.67 / 35.65   | 12       | 36        |
| 8      | 1        | 1000    | 7.86 / 12.32  | 19.76 / 29.46   | 12       | 36        |
| 8      | 10       | 1       | 10.99 / 16.66 | 37.17 / 52.74   | 12       | 63        |

| Fields | Versions | Records | Bulk access     | Point access    | Bulk SQL | Point SQL |
|--------|----------|---------|-----------------|-----------------|----------|-----------|
| 8      | 10       | 100     | 9.05 / 12.07    | 31.33 / 42.47   | 12       | 63        |
| 8      | 10       | 1000    | 11.00 / 15.05   | 37.47 / 48.33   | 12       | 63        |
| 8      | 100      | 1       | 28.09 / 37.51   | 169.89 / 229.18 | 12       | 333       |
| 8      | 100      | 100     | 29.48 / 38.74   | 178.35 / 206.82 | 12       | 333       |
| 8      | 100      | 1000    | 32.39 / 48.33   | 196.73 / 247.61 | 12       | 333       |
| 32     | 1        | 1       | 10.95 / 15.00   | 52.95 / 66.65   | 12       | 108       |
| 32     | 1        | 100     | 12.95 / 19.41   | 61.33 / 77.08   | 12       | 108       |
| 32     | 1        | 1000    | 11.48 / 14.61   | 54.53 / 73.08   | 12       | 108       |
| 32     | 10       | 1       | 16.16 / 22.78   | 77.03 / 98.56   | 12       | 135       |
| 32     | 10       | 100     | 14.55 / 18.50   | 67.70 / 80.59   | 12       | 135       |
| 32     | 10       | 1000    | 14.55 / 19.94   | 68.61 / 85.06   | 12       | 135       |
| 32     | 100      | 1       | 56.13 / 70.49   | 245.75 / 298.96 | 12       | 405       |
| 32     | 100      | 100     | 56.33 / 76.65   | 247.93 / 299.68 | 12       | 405       |
| 32     | 100      | 1000    | 55.65 / 74.35   | 241.16 / 303.78 | 12       | 405       |
| 128    | 1        | 1       | 30.94 / 43.03   | 221.63 / 290.86 | 12       | 396       |
| 128    | 1        | 100     | 28.20 / 35.46   | 199.90 / 256.49 | 12       | 396       |
| 128    | 1        | 1000    | 28.68 / 43.26   | 209.81 / 256.03 | 12       | 396       |
| 128    | 10       | 1       | 42.55 / 61.87   | 245.20 / 304.18 | 12       | 423       |
| 128    | 10       | 100     | 38.42 / 48.50   | 221.52 / 255.50 | 12       | 423       |
| 128    | 10       | 1000    | 39.69 / 48.16   | 233.82 / 285.43 | 12       | 423       |
| 128    | 100      | 1       | 155.23 / 206.20 | 476.07 / 581.57 | 12       | 693       |
| 128    | 100      | 100     | 141.00 / 176.70 | 436.33 / 489.88 | 12       | 693       |
| 128    | 100      | 1000    | 170.24 / 216.75 | 509.09 / 595.85 | 12       | 693       |

At 128 fields, 100 versions, and 1,000 total records, the medians are 170.24 ms for bulk access and 509.09 ms for point access, a point/bulk ratio of 2.99. Statement counts are 12 and 693. This comparison is not a rerun of the historical preoptimization algorithm and is not combined with historical-machine timings. The archive retains paired differences and bootstrap intervals; those intervals describe repeated observations on this machine, not cross-deployment variation.

#### *S4.3. Storage-population boundary*

At 1,000, 10,000, and 100,000 transitions, the full-schema database sizes are 1.52, 13.16, and 134.53 MiB; simplified storage uses 0.48, 2.88, and 28.45 MiB. Incremental bytes per transition are 1,085.4, 1,077.2, and 1,112.3. MiB uses  $2^{20}$  bytes. Rows are directly populated using a synthetic single-field

history with stable evidence references, then checkpointed. They are not evidence that every populated row passed admission. The simplified schema has fewer audit obligations and is not a functionally equivalent alternative. External evidence and raw model-output storage are excluded.

## S5. Historical agent integration and annotation

### *S5.1. Distinct evidence and denominators*

The historical restricted-agent benchmark has 1,260 planned executions across three requested configurations, each with 420 runs. The configuration mapping is `G1 = gpt-5.6-luna`, `G2 = gpt-5.6-terra`, and `D1 = deepseek-v4-flash`; these archived requested names identify recorded requests, not immutable model weights or new calls made for this revision. A prior pinned-plugin fixture has ten passing integration cases, and six AI-origin lifecycle cases match their declared outcomes. These support exercised interface behavior rather than current provider comparisons.

Among 360 planned benign runs, 320 satisfy the frozen strict trajectory and 331 the post-hoc terminal-endpoint rule. Twenty-five lack a scorable behavior verdict, leaving 335 evaluable runs. The eleven-case difference consists of extra proposal or parent-verification calls that violate the trajectory criterion while reaching the terminal result. It measures sensitivity to the scoring construct. The strict yield is 320/360 (88.89%); conditional strict completion is 320/335 (95.52%). The endpoint counterparts are 331/360 (91.94%) and 331/335 (98.81%). Missing cases are retained, not silently excluded from planned-run yields.

Across all scenarios there are 93 recorded runtime failures: 76/420 for D1, 4/420 for G1, and 13/420 for G2. Some failures retain an authority verdict, and three harness failures retain a behavior verdict; evaluability follows available evidence rather than success status alone.

Separately, no unauthorized mutation is recorded in 720 fixed invalid-tuple admission calls or 179 capability-unavailable branches. The latter do not call admission. Their sum is accounting, not 899 independently sampled attacks. Zero violations in this constructed workload do not establish a population failure bound or general prompt-injection security. The restricted interface leaves admission with the trusted host.

Table S9: Benign B1–B4 strict-trajectory completion by configuration. Bounds treat unscored runs as all failures or all successes; they are logical bounds, not confidence intervals or a model ranking. Configuration identifiers are defined in Supplement S5.

| Config. | Planned | Unscored | Strict/evaluable | Rate   | Bound        |
|---------|---------|----------|------------------|--------|--------------|
| G1      | 120     | 1        | 114/119          | 95.80% | 95.00–95.83% |
| G2      | 120     | 12       | 105/108          | 97.22% | 87.50–97.50% |
| D1      | 120     | 12       | 101/108          | 93.52% | 84.17–94.17% |
| All     | 360     | 25       | 320/335          | 95.52% | 88.89–95.83% |

Table S10: Runtime-failure terminal classes crossed with endpoint evaluability.

| Terminal class    | Total | D1 | G1 | G2 | Authority-<br>evaluable | Behavior-<br>evaluable |
|-------------------|-------|----|----|----|-------------------------|------------------------|
| MODEL_API_FAILURE | 44    | 35 | 0  | 9  | 27                      | 0                      |
| INVALID_OUTPUT    | 37    | 37 | 0  | 0  | 37                      | 0                      |
| HARNESS_FAILURE   | 11    | 4  | 4  | 3  | 0                       | 3                      |
| TIMEOUT           | 1     | 0  | 0  | 1  | 0                       | 0                      |
| Total             | 93    | 76 | 4  | 13 | 64                      | 3                      |

### S5.2. Human annotation and unresolved disagreements

One author (X.W.) and one non-author volunteer coded benign terminal completion. They agree on 359/360 original labels (99.72%; three-category Cohen’s  $\kappa = 0.974$ ). Both label 331/335 behavior-evaluable cases as terminally complete. L.H. separately reviews the eleven strict-negative/endpoint-positive cases and retains their strict-negative classification; this is not a replacement of the original completion labels. The separate T153 inter-human discrepancy is adjudicated in the historical record. Author involvement and residual textual cues limit independence and blinding.

A different textual-acknowledgment audit has 325 outputs labeled by X.W. Agreement with repaired rule v2.1 is 286/325 (88.00%;  $\kappa = 0.644$ ). A3 contributes 32 of 39 disagreements and lacks a uniquely specified target field. Those 39 cases have no completed human adjudication; later unblinded AI proposals do not replace their labels. This audit measures human–rule agreement, not inter-human reliability or factual correctness. Complete labels, rationales, confusion matrices, and resampling sensitivity remain in the pinned historical archive. Neither annotation exercise is part of E2’s scripted-browser denominator.

## S6. Additional protocol and scoring details

This section records details beyond the main protocol. Supplement S1 gives the complete E1/E2 input and case lists, environment, execution commands, rejection-audit baseline, and source receipts. S4 gives all timing and storage cells; S5 gives historical integration endpoints. Those records retain their original denominators.

### *S6.1. Comparator and query observations*

The event journal is a separately written `sqlite3` comparator developed for this study, rather than an externally deployed system or an independent-team replication. Both configurations retain immutable candidates and events, review context, version checks, transactional updates, and complete field sources. Context-level authorization retains field, version, evidence, producer, and proposal context while excluding certificate identifier, internal candidate identifier, and instance creation time. Exact authorization adds the reviewed-candidate identity. The reference arm invokes the actual transactional admission service.

The instance policy requires the authorized candidate instance. The value/context policy permits another instance only when the retained observations agree. Evidence substitutions change constructed metadata rather than image contents. The archived inputs are deterministically selected without using task-success labels. Their twelve configuration–scenario cells contain four distinct values, and three synthetic controls add the type boundaries listed in S1.

An independent reader scores six queries for each of three fields. Answers are correct, wrong, ambiguous, or unavailable. Rejection requires the expected constraint violation or injected interruption as well as an unchanged logical fact-database digest; an arbitrary exception is not a successful rejection. The baseline follows authorization preparation, so recorded review intent can predate a rejected admission. The service-generated reference bundle is captured before submission to the transaction. The strengthened-baseline continuity rerun remains outside the 495-execution denominator.

### *S6.2. Browser fixture*

An independent Playwright driver reads the displayed candidate identifier, proposal, and authorized value from the DOM before the authorization click. The original `ReviewForms.confirm` service and experimental server-held

session gate process the same declared interactions. Six initial confirmations prepare replay cases and are setup operations beyond the sixty scored cases. The decimal input avoids JavaScript normalization of an integral float. The fixture uses fixed trusted principal labels; its session database is separate from the fact database.

### *S6.3. Cost measurement boundaries*

Each timing cell uses five warmups and 200 observations per arm, randomly interleaved with seed 20260923. Warmups, database copying, and adapter construction are excluded. The exact journal opens its connection during construction; the reference ORM can obtain its first connection during confirmation. SQL hooks also count trigger execution differently, so absolute cross-implementation statement totals are not compared.

For the separate manual-source materialization ablation, full admission generates a valid relational delta outside the reduced-arm timer. Applying that delta retains complete snapshots, source maps, authority rows, transactionality, and CAS. The resulting state must match the precomputed relational fingerprint. Random review identifiers and timestamps prevent a byte-identical database claim across separate full admissions. This ablation and the recognition-source mechanism comparison use different workloads.

The point-read trace arm changes only the certificate, certificate-evidence, and transition repository reads. Valid unrelated forms are seeded once and copied outside timing. The complete returned trace object must equal its counterpart after every invocation. Storage fixtures use single-field version growth and stable evidence references, with WAL/NORMAL and checkpointing before footprint measurement. Direct schema population measures storage, rather than admission of every populated row.

### *S6.4. Historical scoring*

The strict benign criterion requires the specified exact call sequence; its stale scenario also requires the word **stale**. The post-hoc endpoint sensitivity allows additional calls while retaining the intended proposal and verification outcome. Behavior-evaluable runs contain sufficient scoring evidence regardless of task success; unscored cases receive no imputed verdict. Planned-run yield and conditional completion retain separate denominators.

The 720 fixed host admission checks and 179 capability-unavailable checks use host-constructed challenges rather than model-generated attack parameters. The latter branch makes no admission call. Unauthorized mutation

means failing an expected rejection or changing the authoritative-state digest. Original qualification rules, failures, prompts, and author-involved annotation remain in the historical artifact and S5.

## S7. Extended validity details

The main discussion states the approval-policy choice, trusted review boundary, formal scope, and cost interpretation. This section preserves additional boundaries for the archived evidence.

### *S7.1. Observation and implementation scope*

An omitted observation class cannot be reconstructed by decoding an opaque candidate handle. Other observation models may combine or derive distinctions differently. The paired constructions establish conditional class-wise irredundancy, rather than universal minimality or completeness over every admission failure. Several Alloy assertions check immediate consequences of definitions; selected projection instances and mutants test the abstraction interface, rather than a refinement theorem over every execution.

Canonical JSON equality governs change detection, authorized and committed values, and tracing. The repaired integer/float and numeric/Boolean cases show why these checks need the same value-equivalence relation. Targeted regressions cover selected boundaries, rather than every serialization or migration behavior. The concrete prefix trace exceeds the formal single-step anchor check in some respects; same-value admission and schema uniqueness also differ as stated in the main conformance section.

Evidence checks compare stored digests and canonical locators without re-reading external bytes. Replacing external evidence while retaining its metadata therefore needs immutable storage or another integrity check. Authentication, serialization, policy state, hashing, and transaction execution remain trusted. A revocation visible before the commit-entry policy check causes rejection; a later revocation does not cancel that in-flight transaction. Deletion, redaction, tombstones, retention expiry, distributed revocation, and other database engines require further design and evidence.

### *S7.2. Constructed and historical studies*

The 495 E1 executions are factorial constructed cases using seven distinct proposal values, rather than independent applications. The comparator is

a same-study implementation and evidence contexts are constructed meta-data. E2 is an instrumented local fixture, rather than the deployed product interface or a human-subject study. Its counts describe exercised interactions; authentication, attention, hostile-host resistance, and crash atomicity across the session and fact databases remain unevaluated. The session gate is an experimental extension, not a property attributed to the original path.

Historical hosted-agent runs have separate scoring criteria, missing runtime outcomes, fixed host checks, and author-involved annotation. Fixed challenge tuples measure the exercised integration checks rather than autonomous attack efficacy or population failure probabilities. E1–E3 make no hosted-model calls.

### *S7.3. Measurement interpretation*

Admission timings include differences in schema, ORM, certificate work, application features, and connection lifecycle. The prevalidated-materialization arm moves validation and planning outside its timer and cannot replace full admission safely. The trace experiment changes access within the same current verifier. Twelve SQL statements across the grid do not imply constant total cost: transferred rows, indexes built in memory, and predecessor traversal still grow with history and field count.

One-machine interleaving limits order effects while retaining scheduling, caching, and connection differences. Repetition-based intervals describe that run, rather than uncertainty across machines or deployments. Storage excludes external evidence bytes, model-output files, and production retention policies; the reduced schema has fewer audit obligations. An independent application with operational reviewers, records, and policies would provide the next test of transferability.